

String and Text Manipulation – Interview Questions

1. Remove Spaces from Given Text

Definition:

Remove all spaces from a string.

Program:

```
text = "he llo wor ld"
```

```
result = text.replace(" ", "")
```

```
print(result)
```

Output:

```
helloworld
```

Explanation:

replace(" ", "") replaces every space with nothing.

2. Reverse a String

Definition:

Reverse the characters of a string.

Program:

```
text = "hello"
```

```
result = text[::-1]
```

```
print(result)
```

Output:

```
olleh
```

Explanation:

[::-1] reads the string from the last character to the first character.

3. Reverse a String After Removing Spaces

Definition:

First remove spaces and then reverse the string.

Program:

```
text = "he llo world"
```

```
text = text.replace(" ", "")
```

```
result = text[::-1]
```

```
print(result)
```

Output:

```
dlrowolleh
```

Explanation:

First "he llo world" becomes "helloworld".

Then [::-1] reverses it.

4. Convert Snake Case to Camel Case

Definition:

Convert snake_case into camelCase.

Program:

```
text = "my_variable_name"
```

```
words = text.split("_")
```

```
result = words[0]
```

```
for word in words[1:]:
```

```
    result = result + word.capitalize()
```

```
print(result)
```

Output:

myVariableName

Explanation:

split("_") gives:

```
['my', 'variable', 'name']
```

First word remains lowercase.

Other words are capitalized and joined.

5. Convert Snake Case to Pascal Case

Definition:

Convert snake_case into PascalCase.

Program:

```
text = "my_variable_name"
```

```
words = text.split("_")
```

```
result = ""
```

```
for word in words:
```

```
    result = result + word.capitalize()
```

```
print(result)
```

Output:

MyVariableName

Explanation:

Every word is capitalized and joined together.

6. Convert Camel Case to Snake Case

Definition:

Convert camelCase into snake_case.

Program:

```
text = "myVariableName"
```

```
result = ""
```

```
for char in text:
```

```
    if char.isupper():
```

```
        result = result + "_" + char.lower()
```

```
    else:
```

```
        result = result + char
```

```
print(result)
```

Output:

```
my_variable_name
```

Explanation:

Whenever an uppercase character is found, _ is added before it and the character is converted to lowercase.

7. Convert Camel Case to Pascal Case**Definition:**

Convert camelCase into PascalCase.

Program:

```
text = "myVariable"
```

```
result = text[0].upper() + text[1:]
```

```
print(result)
```

Output:

MyVariable

Explanation:

text[0] gets the first character.

.upper() converts m to M.

8. Convert Pascal Case to Camel Case**Definition:**

Convert PascalCase into camelCase.

Program:

```
text = "MyVariable"
```

```
result = text[0].lower() + text[1:]
```

```
print(result)
```

Output:

myVariable

Explanation:

Only the first character is converted to lowercase.

9. Convert Pascal Case to Snake Case**Definition:**

Convert PascalCase into snake_case.

Program:

```
text = "MyVariable"
```

```
result = ""
```

```
for char in text:
```

```
if char.isupper() and result != "":  
    result = result + "_"  
result = result + char.lower()
```

```
print(result)
```

Output:

```
my_variable
```

Explanation:

When an uppercase character appears in the middle, _ is added before it.

10. Convert Text to Camel Case

Definition:

Convert a space-separated sentence into camelCase.

Program:

```
text = "hello world example"
```

```
words = text.split()
```

```
result = words[0]
```

```
for word in words[1:]:
```

```
    result = result + word.capitalize()
```

```
print(result)
```

Output:

```
helloWorldExample
```

Explanation:

First word stays lowercase.

Remaining words are capitalized.

11. Convert Text to Snake Case

Definition:

Convert a space-separated sentence into snake_case.

Program:

```
text = "hello world example"
```

```
words = text.split()
```

```
result = "_".join(words)
```

```
print(result)
```

Output:

```
hello_world_example
```

Explanation:

join() joins all words using _.

12. Convert Text to Pascal Case

Definition:

Convert a space-separated sentence into PascalCase.

Program:

```
text = "hello world example"
```

```
words = text.split()
```

```
result = ""
```

```
for word in words:
```

```
    result = result + word.capitalize()
```

```
print(result)
```

Output:

HelloWorldExample

Explanation:

Each word is capitalized and joined without spaces.

13. Swap Upper and Lower Case

Definition:

Convert uppercase letters to lowercase and lowercase letters to uppercase.

Program:

```
text = "HeLLo"
```

```
print(text.swapcase())
```

Output:

hELLO

Explanation:

swapcase() changes uppercase → lowercase and lowercase → uppercase.

14. Separate Digits from Text

Definition:

Extract only the digits from an alphanumeric string.

Program:

```
text = "abc123d4"
```

```
result = ""
```

```
for char in text:
```

```
    if char.isdigit():
```

```
result = result + char
```

```
print(result)
```

Output:

1234

Explanation:

isdigit() checks whether the character is a digit.

Only digits are added to result.

15. Print Uppercase, Lowercase, Digits, and Special Characters Separately**Definition:**

Separate characters into uppercase, lowercase, digits, and special characters.

Program:

```
text = "Abc123!@#"
```

```
upper = ""
```

```
lower = ""
```

```
digits = ""
```

```
special = ""
```

```
for char in text:
```

```
    if char.isupper():
```

```
        upper = upper + char
```

```
    elif char.islower():
```

```
        lower = lower + char
```

```
    elif char.isdigit():
```

```
        digits = digits + char
```

```
    else:
```

```
        special = special + char
```

```
print("Uppercase:", upper)
print("Lowercase:", lower)
print("Digits:", digits)
print("Special Characters:", special)
```

Output:

Uppercase: A

Lowercase: bc

Digits: 123

Special Characters: !@#

Explanation:

- isupper() → uppercase
 - islower() → lowercase
 - isdigit() → digits
 - else → special characters
-

16. Count Uppercase, Lowercase, Digits, and Special Characters**Definition:**

Count how many uppercase, lowercase, digits, and special characters are present.

Program:

```
text = "AbC@123x!"
```

```
upper = 0
```

```
lower = 0
```

```
digits = 0
```

```
special = 0
```

```
for char in text:
```

```
if char.isupper():
    upper = upper + 1
elif char.islower():
    lower = lower + 1
elif char.isdigit():
    digits = digits + 1
else:
    special = special + 1

print("Uppercase:", upper)
print("Lowercase:", lower)
print("Digits:", digits)
print("Special Characters:", special)
```

Output:

Uppercase: 2

Lowercase: 2

Digits: 3

Special Characters: 2

Note:

For the given input "AbC@123x!", lowercase characters are b and x, so the correct lowercase count is **2**.

17. Check Password Strength

Definition:

Check whether a password contains at least one uppercase, one lowercase, one digit, and one special character.

Program:

```
password = "Pass123!"
```

```
upper = False
```

```
lower = False
```

```
digit = False
```

```
special = False
```

```
for char in password:
```

```
    if char.isupper():
```

```
        upper = True
```

```
    elif char.islower():
```

```
        lower = True
```

```
    elif char.isdigit():
```

```
        digit = True
```

```
    else:
```

```
        special = True
```

```
if upper and lower and digit and special:
```

```
    print("Strong Password")
```

```
else:
```

```
    print("Weak Password")
```

Output:

Strong Password

Explanation:

The password has:

- P → uppercase
- ass → lowercase
- 123 → digits
- ! → special character

So it is a strong password.

18. Remove Duplicates in a Given Input

Definition:

Remove repeated characters and keep only the first occurrence.

Program:

```
text = "aabbcc"
```

```
result = ""
```

```
for char in text:
```

```
    if char not in result:
```

```
        result = result + char
```

```
print(result)
```

Output:

```
abc
```

Explanation:

char not in result checks whether the character is already present.

19. Print Duplicates in a Given String

Definition:

Find characters that occur more than once.

Program:

```
text = "aabbccde"
```

```
result = ""
```

```
for char in text:
```

```
    if text.count(char) > 1 and char not in result:
```

```
result = result + char
```

```
print(result)
```

Output:

abc

Explanation:

text.count(char) counts how many times each character occurs.

If the count is greater than 1, it is a duplicate.

20. Print Next Characters in a Given String**Definition:**

Replace every character with the next character in the alphabet.

Program:

```
text = "abc"
```

```
result = ""
```

```
for char in text:
```

```
    result = result + chr(ord(char) + 1)
```

```
print(result)
```

Output:

bcd

Explanation:

For a:

ord("a") = 97

97 + 1 = 98

chr(98) = b

Similarly:

$a \rightarrow b$

$b \rightarrow c$

$c \rightarrow d$

So the output is:

bcd